Learning Objectives

After studying this chapter, the students will be able to understand

- the indefinite integral.
- how to find the indefinite integral of a function involving sum, difference and constant multiples.
- how to use and where to apply the substitution technique in indefinite integrals.
- the techniques involved in integration by parts and some special type of integrals.
- the fundamental theorems of integral calculus.
- the properties of definite integral and its applications.
- the application of a particular case of gamma integral.
- the properties of gamma function.
- the evaluation of the definite integral as the limit of a sum.

2.1 Indefinite Integrals

2.1.1 Concept of Indefinite Integral

In differential calculus, we have learned how to calculate the differential coefficient $f'(x)$ of a given function $f(x)$ with respect to x. In this chapter, we have to find out the primitive function $f(x)$ (i.e. original function) whenever its derived function $f'(x)$ (i.e. derivative of a function) is given, such process is called integration or anti differentiation.

∴ Integration is the reverse process of differentiation

We know that $\frac{d}{dx}(\sin x) = \cos x$. Here $\cos x$ is known as **Derived function**, and $\sin x$ is known as **Primitive function** [also called as Anti derivative function (or) Integral function].

Definition 2.1

A function $F(x)$ is said to be a primitive function of the derived function $f(x)$, if $\frac{d}{dx}\left[F(x)\right] = f(x)$

Now, consider the following examples which are already known to us.

$\frac{d}{dx}\left(x^3\right) = 3x^2$, $\frac{d}{dx}\left(x^3+5\right) = 3x^2$,

$\frac{d}{dx}\left(x^3-\frac{3}{2}\right) = 3x^2$, $\frac{d}{dx}\left(x^3+e\right) = 3x^2$,

$\frac{d}{dx}\left(x^3-\pi\right) = 3x^2$, ...

From the above examples, we observe that $3x^2$ is the derived function of the primitive functions x^3, x^3+5, $x^3-\frac{3}{2}$, x^3+e, $x^3-\pi$, ... and which indicates that the primitive functions are need not be unique, even though the derived function is unique. So we come to a conclusion that x^3+c is the primitive function of the derived function $3x^2$.

∴ For every derived function, there are infinitely many primitives by choosing c arbitrarily from the set of real numbers $\mathbb{R}$. So we called these integrals as indefinite integrals.

In general,

$$\frac{d}{dx}\left[F(x)\right] = f(x) \Rightarrow \int f(x)\,dx = F(x)+c,$$

where c is called the constant of integration.

Remarks

- If two different primitive functions $F(x)$ and $G(x)$ have the same derived function $f(x)$, then they differ only by a constant term.
- $\int f(x)\,dx$ is called as indefinite integral.

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